

Jayanth Ravimurugan

AI Engineer

+1(608)-515-2600 | jayanthravimurugan@gmail.com | [linkedin.com/in/jayanth-r-x](https://www.linkedin.com/in/jayanth-r-x) | github.com/XELJayX | Madison, WI

EDUCATION

University of Wisconsin–Madison

Master of Science in Data Science

Madison, WI

2025 – 2027

Amity University Haryana

Integrated B.Tech + M.Tech in Artificial Intelligence and Machine Learning

Haryana, India

2020 – 2025

SKILLS

Programming Languages: Python, SQL, R, C++, Java, Kotlin, TypeScript, Bash

AI/ML & LLM Engineering: PyTorch, scikit-learn, HuggingFace Transformers, LangChain, LangGraph, agentic systems, RAG pipelines, prompt engineering, fine-tuning, BERT, LLaMA, GPT, AWS Bedrock

Data & Retrieval: Vector databases (ChromaDB, FAISS), embeddings, semantic search, Pandas, NumPy

MLOps & Cloud Infrastructure: Docker, FastAPI, Celery, Redis, AWS (ECS, EC2, Bedrock), CI/CD (GitHub Actions), REST APIs, Weights & Biases

Full-Stack & Mobile Development: React, TypeScript, Tiptap, PostgreSQL, Kotlin, Jetpack Compose, Android, TorchScript (on-device ML)

EXPERIENCE

Student Research Assistant

N+1 Institute, University of Wisconsin–Madison

2026 – Present

Madison, WI

- Developing a full-stack **Retrieval-Augmented Generation (RAG)** web application for the UW Department of Surgery that helps researchers draft grant applications with AI assistance, using a **React/TypeScript** frontend and a **Python/FastAPI** backend.
- Integrated a **Tiptap**-based rich-text editor (Google Docs-style) enabling researchers to co-write grant applications with real-time AI assistance.
- Built document ingestion, chunking, embedding, and vector-retrieval pipelines integrated with **AWS Bedrock** LLM infrastructure; deployed on **AWS ECS/EC2** with **CI/CD** automation while handling **HIPAA**-protected data.

Data Science Intern

OneBanc Technologies

Nov 2024 – Feb 2025

Gurugram, India

- Collected and cleaned **~500K** raw SMS records via a custom **Python** ETL pipeline, producing a fully annotated **NER** corpus used across all downstream models.
- Engineered a **custom financial tokenizer** preserving domain entities (account numbers, amounts, merchant names), reducing out-of-vocabulary tokens by **~30%** versus standard tokenizers.
- Benchmarked **5 architectures** (**CRF, LSTM, BiLSTM, ANN, BERT**) for financial **NER**; fine-tuned **BERT** achieved **92.3% F1**, outperforming the BiLSTM baseline by **~8 pp**.
- Wrote **Bash/PowerShell** automation scripts and **YAML CI/CD** pipelines, and refactored a **C#** codebase, reducing manual deployment steps by **~40%**.

PROJECTS

AutoReviewer — Autonomous AI Code Review Agent | github.com/XELJayX/PRCodeReviewer

- Built a fully autonomous AI agent that reviews **GitHub** pull requests in real time using a **LangGraph** multi-step agent pipeline (**Groq LLaMA 3.3 70B**), surfacing security, performance, and quality issues across **4 review dimensions**.
- Designed an event-driven architecture (**FastAPI + Celery + Redis**) for asynchronous webhook processing; implemented **HMAC-SHA256** validation and cut memory footprint by **~95%** to fit a **512 MB** deployment budget.

Mini LLM — Decoder-Only Transformer Built From Scratch | [Live Demo](#) | [Repo](#)

- Implemented a **decoder-only transformer** (273K parameters) from scratch in **PyTorch**, including a custom **BPE tokenizer** (the same algorithm used by GPT-4), multi-head causal self-attention, positional embeddings, and residual connections, with no HuggingFace model libraries used.
- Built a production-grade training pipeline with **AdamW**, cosine learning-rate scheduling, gradient clipping, dropout regularization, and early stopping (best validation loss **3.07** at epoch 7), with experiment tracking via **Weights & Biases**.
- Deployed full-stack: **FastAPI** inference backend and **React** chat UI, containerized with **Docker** and hosted on **Hugging Face Spaces**.

Text-to-SQL System with Schema-Aware RAG Pipeline | [Live Demo](#)

- Built a production **Text-to-SQL** system with a schema-aware **RAG** pipeline (**fastembed + ChromaDB**) for dynamic table retrieval, eliminating context-window bloat on schemas with **50+ tables**.
- Implemented a **SQL validation** and self-correction loop via the **Groq Llama 3.1 API**; evaluation over **17 test cases** achieved **100%** accuracy on simple queries and **60%** on medium/hard queries, on par with published baselines for schema-aware systems; deployed full-stack (**FastAPI/React/PostgreSQL**) on Railway and Vercel.

PUBLICATIONS

On-Device Financial Named Entity Recognition Using Deep Learning — First Author, ICDISS 2025 (IEEE)

2025

- Designed and benchmarked **5 NER architectures** for on-device deployment on resource-constrained hardware; proposed a quantization and pruning strategy achieving **60%** model size reduction with under **2 pp** of F1 degradation.